

Solution Architecture

Date: 15 March 2026

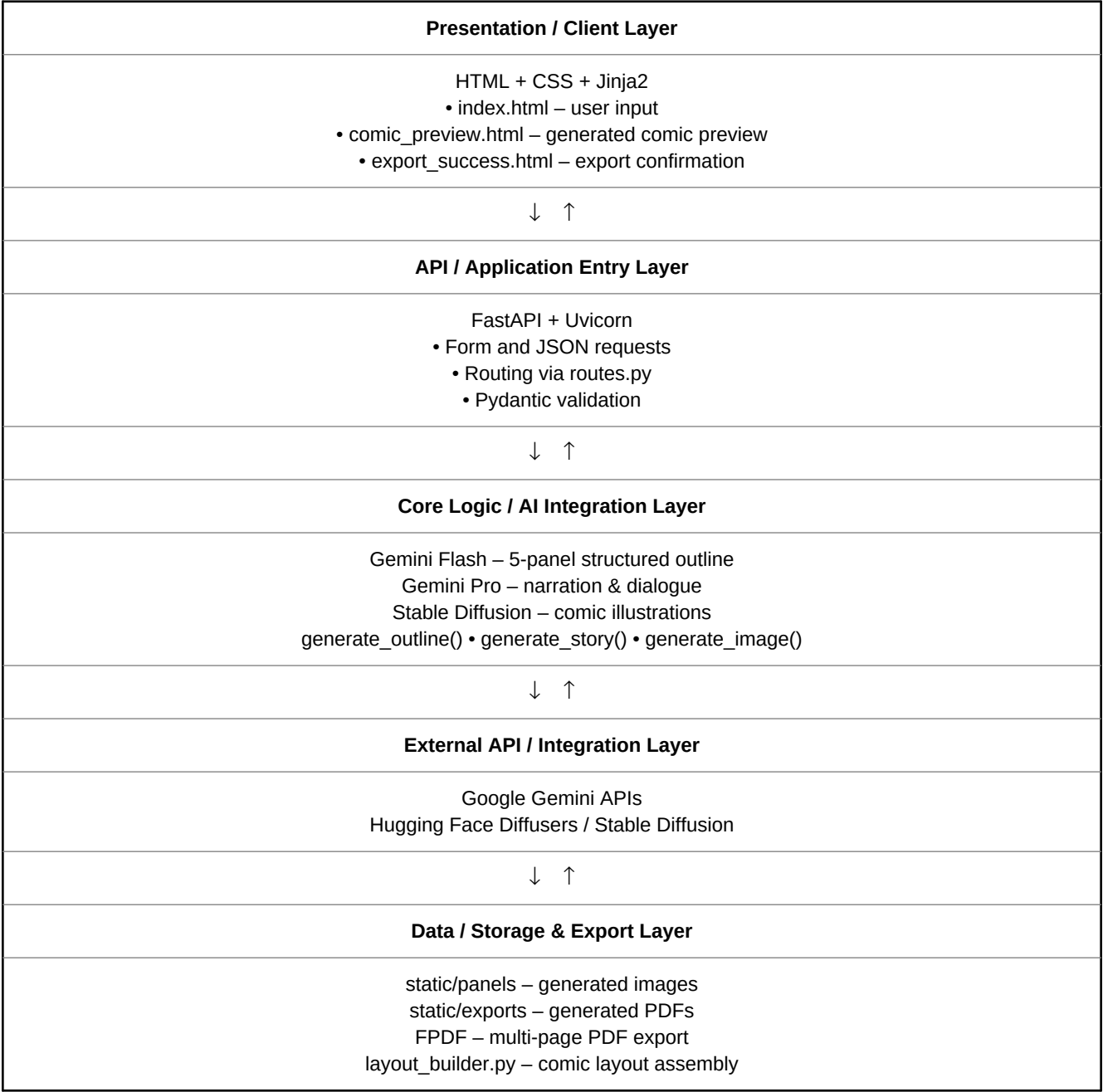
Team ID: SWTID-2026-7446

Project Name: ComicCraft - AI Comic Story Creator using Gemini Models

Maximum Marks: 5 Marks

Solution Architecture Diagram

High-level architecture mapped to the provided template using the technologies and components described in the ComicCraft project document.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects story prompt, character name, setting, tone and art style; displays generated comic panels and export status.	HTML, CSS, Jinja2
API / Application Layer	Receives form/JSON input, validates requests, manages routing and coordinates the comic generation workflow.	FastAPI, Uvicorn, Pydantic
Core Logic / AI Services	Generates a structured 5-panel outline, expands it into narration/dialogue, creates illustrations, and builds the final comic layout.	Gemini Flash, Gemini Pro, Stable Diffusion, Python modules
External API Integrations	Provides AI generation services for text/story content and comic-style image creation.	Google Gemini APIs, Hugging Face Diffusers
Data / Storage & Export	Stores generated panel images and exported PDFs; compiles panel images and narration into downloadable multi-page PDFs.	static/panels, static/exports, FPDF, Pillow

Architecture Flow: User Input → FastAPI → Gemini Flash → Gemini Pro → Stable Diffusion → Layout Builder → FPDF Export → Downloadable Comic PDF.

Source basis: ComicCraft project document — technical architecture, AI integration points, backend responsibilities, and export workflow.